

1. Explain a problem solving agent with algorithm.

A problem-solving agent is an intelligent agent that tries to reach a goal by formulating and solving problems. It operates by searching through possible sequences of actions that will lead it from the current state to a goal state.

Key Components of a Problem-Solving Agent:

1. Initial State: The starting point of the agent.
 2. Actions: A set of all possible actions the agent can take.
 3. Transition Model: Describes what state results from performing an action.
 4. Goal Test: A way to determine if the current state is a goal state.
 5. Path Cost: A function that assigns a numeric cost to a path (optional but used in informed searches).
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Steps in a Problem-Solving Agent:

1. Formulate the Problem – Define initial state, goal, actions, and path cost.
 2. Search for a Solution – Apply a search algorithm to find a path from the initial state to a goal state.
 3. Execute the Solution – Perform the actions found in the solution path.
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General Algorithm for a Problem-Solving Agent:

function PROBLEM-SOLVING-AGENT(percept) returns an action:

static: state, an internal representation of the world
goal, a goal the agent is trying to achieve
problem, formulation of the problem to be solved
plan, sequence of actions to be executed

state $\leftarrow$ UPDATE-STATE(state, percept)

if plan is empty then

goal $\leftarrow$ FORMULATE-GOAL(state)

problem $\leftarrow$ FORMULATE-PROBLEM(state, goal)

```
plan ← SEARCH(problem)
action ← FIRST(plan)
plan ← REST(plan)
return action
```

🌀 Example:

Let's say an agent wants to reach from city Arad to Bucharest (Romania map problem):

- Initial State: Arad
- Goal State: Bucharest
- Actions: Move to a connected city
- Transition Model: Describes the road from one city to another
- Goal Test: Is the current city Bucharest?
- Path Cost: Total distance covered

The agent would use a search algorithm (e.g., BFS, DFS, A*) to find a sequence of moves that leads to the goal.

🧠 Types of Search Algorithms:

- Uninformed (Blind) Search: BFS, DFS, UCS
- Informed (Heuristic) Search: Greedy BFS, A*

Let me know if you want a visual example or code for a specific algorithm like BFS or A*.

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2. Write a note on well-defined problems and solutions.

Note on Well-Defined Problems and Solutions

Well-Defined Problems are problems that have a clear starting point, a specific goal, and a defined set of rules or constraints. These problems are structured in a way that makes it easier to apply algorithms or systematic methods to find a solution.

Characteristics of Well-Defined Problems:

1. **Initial State** – The starting point of the problem is clearly stated.
2. **Goal State** – The desired outcome or solution is well specified.

3. **Available Operators** – The possible actions or steps to transition from the initial state to the goal state are clearly defined.
4. **Constraints** – Rules that limit the possible solutions or paths.
5. **Test for Success** – A method to check if the goal has been reached.

Examples:

- Solving a mathematical equation
 - Playing chess
 - Navigating a maze
-

Well-Defined Solutions are solutions that follow logical steps to reach the goal state from the initial state, respecting all the constraints and using the available operators. These solutions are:

- **Correct** – They achieve the desired result.
- **Complete** – They account for all rules and constraints.
- **Efficient** – They aim to use the least resources or steps where possible.

Importance:

- Well-defined problems and solutions are easier to implement in computer programs.
 - They are crucial in fields like artificial intelligence, mathematics, and engineering.
-

In contrast, **ill-defined problems** lack one or more of these components, making them harder to solve systematically.

3. Explain the 8-puzzle problem

🌸 8-Puzzle Problem Explained

The **8-puzzle problem** is a classic **AI search problem** used to demonstrate problem-solving agents and search algorithms.

🧩 1. Problem Description:

- It consists of a **3×3 grid** with **8 numbered tiles** (1 to 8) and **1 blank space**.
- The goal is to **arrange the tiles in order** from 1 to 8, with the blank tile at the bottom-right corner.

2. Example:

Initial State:

```
1 2 3
4 6
7 5 8
```

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Goal State:

```
1 2 3
4 5 6
7 8
```

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3. Allowed Moves (Actions):

The **blank space** can be moved in four directions (if possible):

- Up
- Down
- Left
- Right

4. Components of 8-Puzzle as a Well-Defined Problem:

Component	Description
Initial State	Starting arrangement of tiles
Actions	Move blank tile up, down, left, or right (within grid limits)
Transition Model	Resulting configuration after a move
Goal Test	Check if tiles are in the order 1 to 8 with blank at the end
Path Cost	Each move has a cost of 1; total cost = number of moves

5. Solution:

- A **solution** is a **sequence of moves** that transforms the initial state into the goal state.
- For example:
Move blank right → Move 6 down → Move 5 right → ...

6. Search Techniques Used:

To solve the 8-puzzle, algorithms like:

- **Breadth-First Search (BFS)**
- **Depth-First Search (DFS)**
- **A* Search (with heuristics like misplaced tiles or Manhattan distance)**

are commonly used.

7. Heuristics (used in informed search):

- **Misplaced Tiles:** Count of tiles not in the correct position.
- **Manhattan Distance:** Sum of the distances of each tile from its goal position (row and column differences).

Let me know if you want the code for solving the 8-puzzle using A* or another algorithm!

4. Explain the 8 queen problem

8-Queens Problem Explained

The 8-Queens Problem is a famous constraint satisfaction problem (CSP) in artificial intelligence and computer science.

Problem Statement:

- Place 8 queens on a standard 8×8 chessboard such that:
 - No two queens attack each other.
 - That means no two queens can be in the same row, column, or diagonal.
-



Key Constraints:

1. Only one queen per row.
2. Only one queen per column.
3. No two queens on the same diagonal.



Why is it important?

- It is a classic example of a constraint satisfaction problem (CSP).
- It demonstrates backtracking, state space search, and constraint checking.



Approach to Solve:



Backtracking Algorithm (most common method):

1. Start placing queens one by one in each row.
2. For each row, try placing a queen in all columns (from left to right).
3. If a safe position is found (no conflict with previously placed queens), place the queen.
4. If no safe position is found, backtrack to the previous row and move the queen to the next column.
5. Repeat until all 8 queens are placed safely.



State Space:

- Each state represents a partially filled board.
- Total possible combinations without constraints: $64C8 \approx 4.4$ million
- With one queen per row: $8^8 = 16,777,216$
- But backtracking significantly reduces the number of checks.



Solution Representation:

You can represent a solution as an array:

csharp

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```
[0, 4, 7, 5, 2, 6, 1, 3]
```

- Index = row number
- Value = column position of the queen in that row

Example: Queen in row 0 is placed at column 0, row 1 at column 4, and so on.

✓ Valid Solution Example (Graphically):

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```
Q . . . . .
. . . . Q . .
. . . . . Q
. . . Q . . .
. . Q . . . .
. . . . . Q .
. Q . . . . .
. . . . . Q .
```

Let me know if you want the code implementation using backtracking in Java, Python, or another language!

5. BFS Algorithm (Breadth-First Search) – Explanation with Example

Breadth-First Search (BFS) is a graph traversal algorithm used to explore nodes and edges of a graph or tree level by level. It starts at a source node and explores all its neighbors before moving on to their neighbors.

Key Characteristics of BFS:

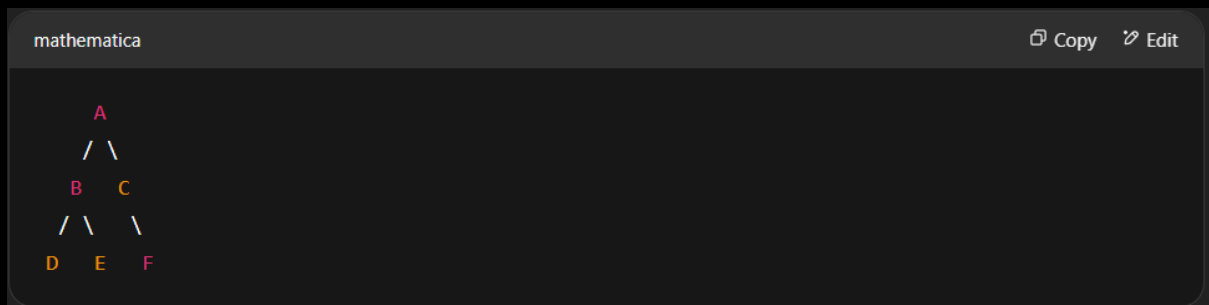
- Uses **Queue (FIFO)** data structure.
- Visits nodes in the order of their **distance from the source**.
- Suitable for finding the **shortest path** in an unweighted graph.

BFS Algorithm Steps:

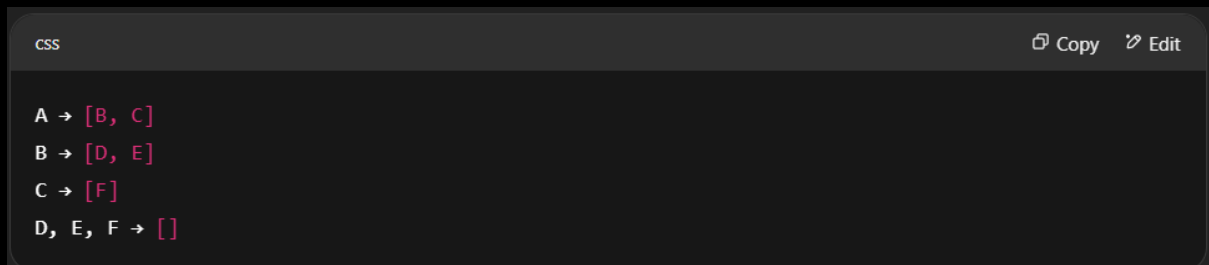
1. Start from the source node and mark it as **visited**.
 2. Add the source node to the **queue**.
 3. While the queue is not empty:
 - Dequeue a node from the front.
 - Visit all its **unvisited neighbors**, mark them as visited, and enqueue them.
 4. Repeat until all reachable nodes are visited.
-

Example:

Graph:



Adjacency List:



BFS Traversal from Node A:

Steps:

Queue	Visited Order
A	A
B, C	A, B
C, D, E	A, B, C
D, E, F	A, B, C, D
E, F	A, B, C, D, E
F	A, B, C, D, E, F

Final BFS Traversal Output:

$A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F$

Applications of BFS:

- Finding the shortest path in an unweighted graph
 - Web crawling
 - GPS navigation systems
 - Social networking suggestions
-

Let me know if you want the BFS implemented in code (C++, Java, Python, etc.).

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6. Uniform Cost Search (UCS) Algorithm – Explanation with Example

Uniform Cost Search (UCS) is a graph traversal algorithm used to find the **least cost path** from a starting node to a goal node. It is an informed search algorithm that expands the **least-cost node** first, regardless of depth.

Key Features of UCS:

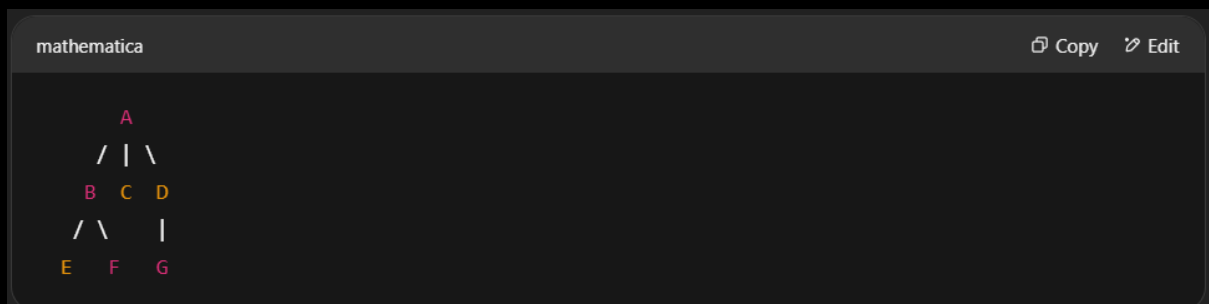
- It uses a **priority queue (min-heap)** to explore the lowest cumulative cost node first.
- It is **complete** (guarantees a solution if one exists).
- It is **optimal** (guarantees the least-cost path).
- It is similar to **Dijkstra's Algorithm**.

✅ Uniform Cost Search Algorithm Steps:

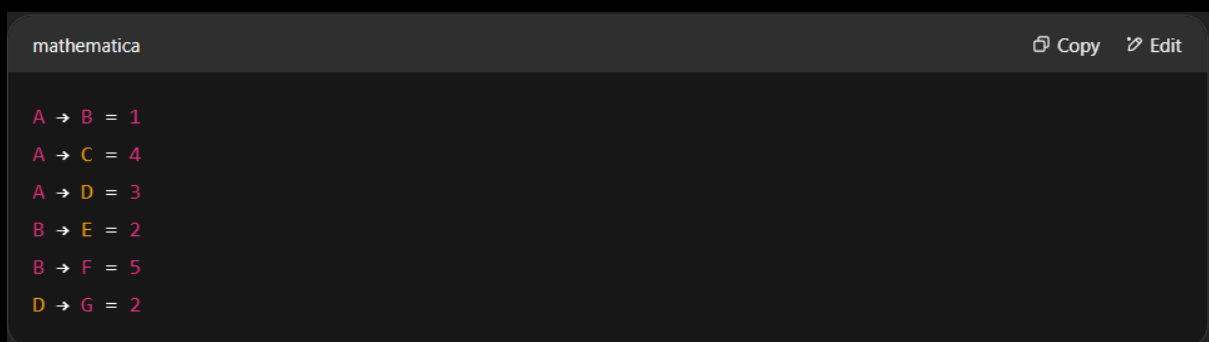
1. Start with the source node and set its cost to 0.
 2. Insert the node into a **priority queue** based on path cost.
 3. Repeat until the queue is empty:
 - Remove the node with the **lowest path cost**.
 - If it is the **goal node**, return the path.
 - Otherwise, expand the node:
 - For each neighbor, calculate cumulative cost.
 - If the neighbor is not visited or found with a lower cost, update and add to the queue.
-

🧠 Example:

Graph:



Edge Costs:



Goal: Find the least-cost path from A to G

🔍 Execution (step-by-step):

Step	Node Explored	Path	Cumulative Cost	Queue (Cost:Node)
1	A	A	0	1:B, 4:C, 3:D
2	B	A→B	1	3:D, 4:C, 3:E, 6:F
3	D	A→D	3	4:C, 3:E, 6:F, 5:G
4	E	A→B→E	3	4:C, 5:G, 6:F
5	C	A→C	4	5:G, 6:F
6	G	A→D→G	5	6:F

✅ Goal node G found with cost 5.

❖ Final Result:

Least-cost path: $A \rightarrow D \rightarrow G$

Total cost: $3 + 2 = 5$

📌 Applications of UCS:

- GPS and map navigation (finding shortest/cheapest route)
 - AI game agents (pathfinding with cost constraints)
 - Robot motion planning
-

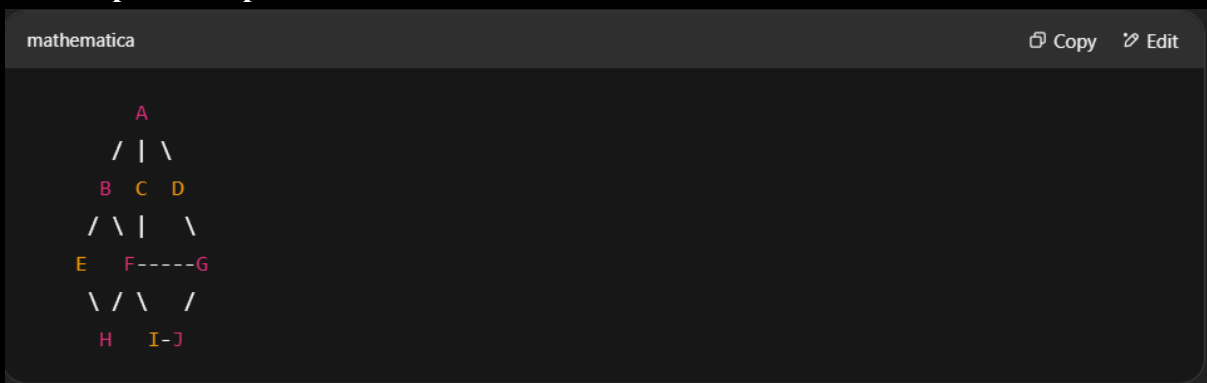
Let me know if you'd like a code implementation in Python, Java, or C++.

ANS 2

but give one complex graph with extra edges which may have multiple paths

Sure! Let's walk through **Uniform Cost Search (UCS)** with a **complex graph** that includes **multiple paths and extra edges**, to understand how UCS finds the **least-cost path** accurately.

Graph Description:



Edges with Costs:

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$A \rightarrow B = 2$	$A \rightarrow C = 3$	$A \rightarrow D = 5$
$B \rightarrow E = 4$	$B \rightarrow F = 2$	
$C \rightarrow F = 1$		
$F \rightarrow G = 5$	$F \rightarrow I = 8$	
$E \rightarrow H = 3$		
$H \rightarrow I = 2$		
$G \rightarrow J = 2$	$I \rightarrow J = 1$	
$D \rightarrow G = 2$		

Goal: Find the least-cost path from A to J

Step-by-Step UCS Execution:

We'll use a **priority queue** to always explore the lowest-cost path first.

Step	Node Explored	Path	Total Cost	Queue (Cost:Path)
1	A	A	0	2:B, 3:C, 5:D
2	B	A→B	2	3:C, 5:D, 6:E, 4:F
3	F	A→B→F	4	3:C, 5:D, 6:E, 9:G, 12:I
4	C	A→C	3	5:D, 6:E, 9:G, 12:I
5	D	A→D	5	6:E, 9:G, 12:I, 7:G (from D)
6	E	A→B→E	6	7:G, 9:G, 12:I, 9:H
7	G	A→D→G	7	9:G, 12:I, 9:H, 9:J
8	J	A→D→G→J	9	9:G, 12:I, 9:H ← goal found

✅ Final Answer:

- **Least-cost path from A to J:** $A \rightarrow D \rightarrow G \rightarrow J$
- **Total cost:** $5 (A \rightarrow D) + 2 (D \rightarrow G) + 2 (G \rightarrow J) = 9$

Even though there were **multiple paths to J** (e.g., via $F \rightarrow I \rightarrow J$ or $E \rightarrow H \rightarrow I \rightarrow J$), UCS correctly found the one with the **lowest total cost** by always expanding the least-cost node first.

💡 Why UCS is Effective in Complex Graphs:

- It doesn't get confused by shorter-looking but **higher-cost** paths.
- It guarantees the **minimum cost**, even if the shortest path in terms of steps isn't the best.

7. DFS (Depth-First Search) Algorithm – Explanation with Example

Depth-First Search (DFS) is a graph traversal algorithm that explores as far as possible along each branch before backtracking. It uses a **stack** (either explicitly or via recursion) to keep track of the path being explored.

🔑 Key Features of DFS:

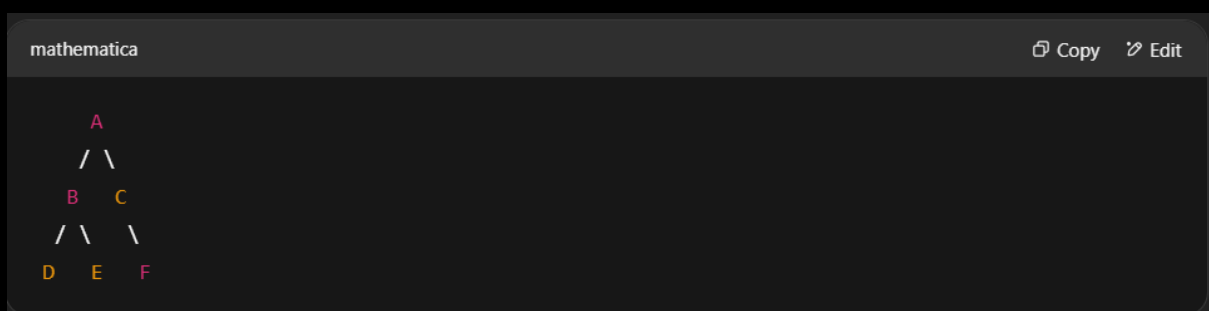
- Explores **deep** before **wide**

- Can be **implemented using recursion** or a **stack**
 - **Does not guarantee** the shortest path
 - Useful in:
 - Solving puzzles (like mazes)
 - Topological sorting
 - Detecting cycles in a graph
-

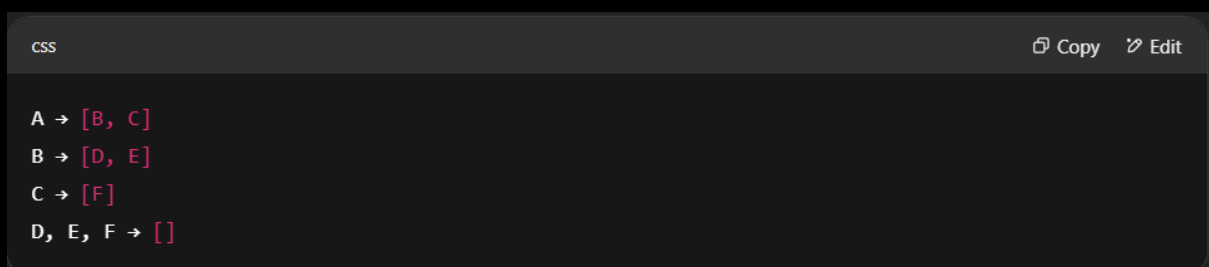
✅ DFS Algorithm Steps:

1. Start at the source node.
 2. Mark it as **visited**.
 3. For each **unvisited neighbor**, **recursively** visit that node.
 4. Backtrack when all neighbors are visited.
-

🧠 Example Graph:



Adjacency List:



📄 DFS Traversal (starting from A):

Let's do DFS using a stack (or recursion):

Step-by-step traversal:

1. Start at A → visited: A
 2. Go to B → visited: A, B
 3. Go to D → visited: A, B, D
(D has no neighbors, backtrack)
 4. Back to B, go to E → visited: A, B, D, E
(E has no neighbors, backtrack)
 5. Back to A, go to C → visited: A, B, D, E, C
 6. Go to F → visited: A, B, D, E, C, F
-

❖ Final DFS Traversal Order:

A → B → D → E → C → F

Note: DFS order **may vary** depending on the order of neighbors in the adjacency list.

✚ Applications of DFS:

- Solving mazes and puzzles
 - Detecting cycles in graphs
 - Topological sorting
 - Pathfinding in AI (less optimal than BFS for shortest path)
-

Let me know if you want DFS implemented in code (Python, C++, Java, etc.) or want to see it in a more complex graph.

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8. Depth-Limited Search (DLS) Algorithm – Explanation with Example

Depth-Limited Search (DLS) is a variation of **Depth-First Search (DFS)** that limits how deep the search can go. It is useful when the search space is large or infinite, and we want to **prevent going too deep**, which may lead to infinite loops or unnecessary computation.

🔑 Key Features of DLS:

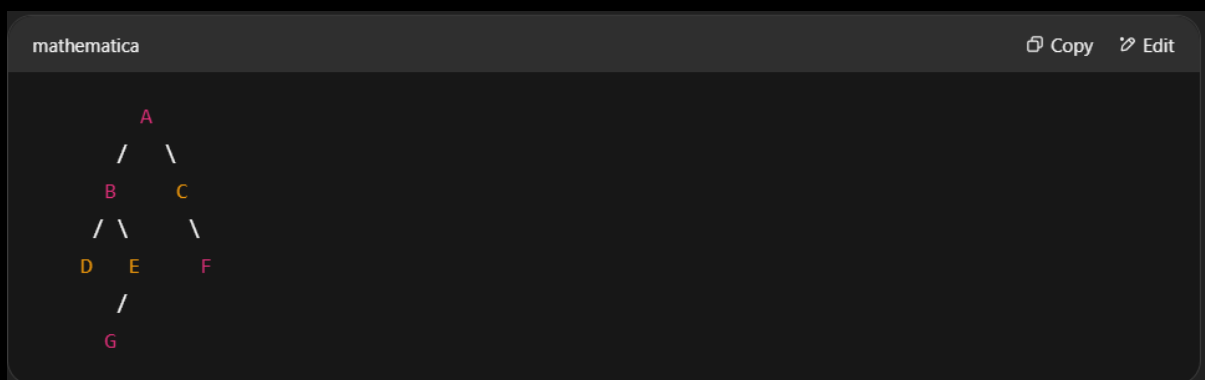
- DFS with a **predefined depth limit**
- Avoids infinite loops in **cyclic or infinite graphs**

- **Incomplete** if the solution is deeper than the limit
 - **More efficient** than DFS when we know the solution is within a certain depth
-

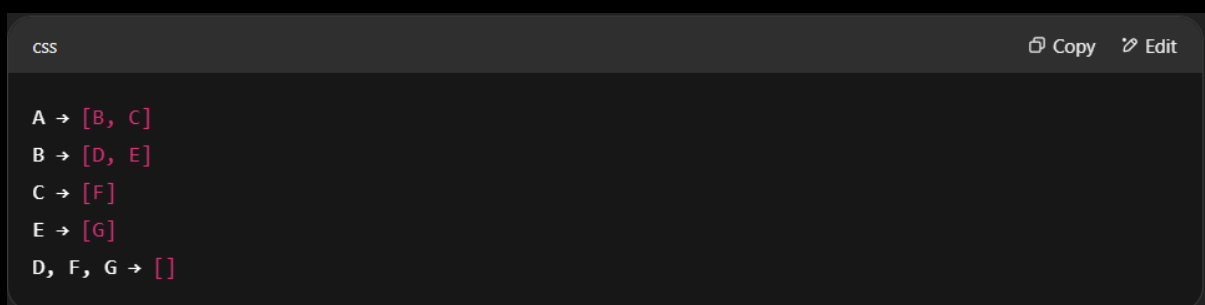
✅ DLS Algorithm Steps:

1. Start at the root (initial) node.
 2. If the node is the goal, return success.
 3. If the current depth = limit, **cut off** further search (do not go deeper).
 4. Otherwise, recursively explore each child, increasing the depth by 1.
 5. If none leads to a solution, return failure.
-

🧠 Example Graph:



Adjacency List:



🎯 Goal: Find node G

🔴 Depth Limit: 2

📄 DLS Traversal with Limit = 2:

- Start at A (depth 0)
- Go to B (depth 1)
 - Go to D (depth 2) → No match, limit reached
 - Go to E (depth 2) → No match, limit reached (G is child of E but we can't go deeper)
- Go to C (depth 1)
 - Go to F (depth 2) → No match

 **No solution found within depth limit 2**

 **Now try with Depth Limit = 3:**

- Start at A
- Go to B
 - Go to E
 - Go to G →  Found at depth 3

 **Final Answer (with depth limit = 3):**

Path: $A \rightarrow B \rightarrow E \rightarrow G$

Depth: 3

 **Limitations of DLS:**

- If depth limit is too **low**, it may miss the solution.
 - If too **high**, behaves like DFS (risking inefficiency or infinite loops).
-

 **Applications of DLS:**

- Ideal for problems with **known depth bounds**
 - Used in **Iterative Deepening Search (IDS)** as a base step
 - Useful in **AI search** when memory is limited
-

Let me know if you'd like a code version or to see how it's used in AI (e.g., game trees).

9. Iterative Deepening Depth-First Search (IDDFS) – Explanation with Example

Iterative Deepening Depth-First Search (IDDFS) combines the space-efficiency of **Depth-First Search (DFS)** and the optimality of **Breadth-First Search (BFS)**.

It repeatedly performs **Depth-Limited Search (DLS)**, increasing the depth limit one level at a time until the goal is found.

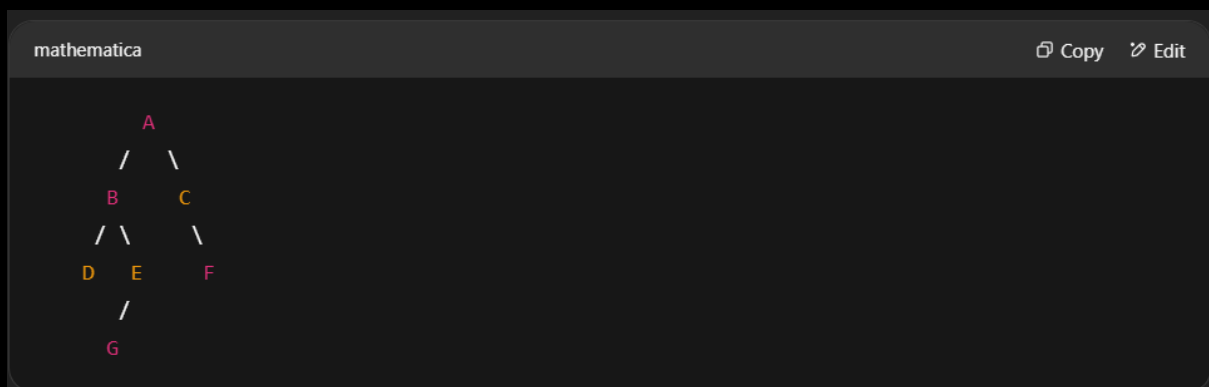
🔑 Key Features of IDDFS:

- Performs DFS up to depth 0, then depth 1, then depth 2, and so on.
- At each level, it **restarts** the search from the root.
- **Complete** and **optimal** (like BFS) when step cost is uniform.
- Uses less memory than BFS (same as DFS).

✅ IDDFS Algorithm Steps:

1. Set depth limit = 0
2. Repeat:
 - Perform **Depth-Limited Search (DLS)** up to the current depth
 - If goal is found, return the path
 - Else, increase depth limit and repeat

🧠 Example Graph:



Adjacency List:

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```
A → [B, C]
B → [D, E]
C → [F]
E → [G]
D, F, G → []
```

Goal: Find node G

IDDFS Execution (Goal = G)

Iteration 1 (depth = 0):

- Visit A
→ Not goal

Iteration 2 (depth = 1):

- Visit A
- Visit B, C
→ Not goal

Iteration 3 (depth = 2):

- Visit A
- Visit B → D, E
- Visit C → F
→ Not goal

Iteration 4 (depth = 3):

- Visit A
- Visit B → D, E
- E → G →  **Goal found**

 **Final Answer:**

Path: A → B → E → G

Depth Found: 3

 **Comparison:**

Feature	DFS	BFS	IDDFS
Complete	No	Yes	Yes
Optimal	No	Yes	Yes
Time	$O(b^d)$	$O(b^d)$	$O(b^d)$
Space	$O(d)$	$O(b^d)$	$O(d)$

✖ Applications of IDDFS:

- AI game trees (like chess or checkers)
- Robot navigation
- Tree-based puzzle solvers

Let me know if you'd like the code for IDDFS in Python or another language.

10. Bidirectional Search Algorithm – Explanation with Example

Bidirectional Search is an efficient graph search algorithm that **simultaneously searches forward from the start node and backward from the goal node**, and stops when the two searches meet in the middle.

🔑 Key Features of Bidirectional Search:

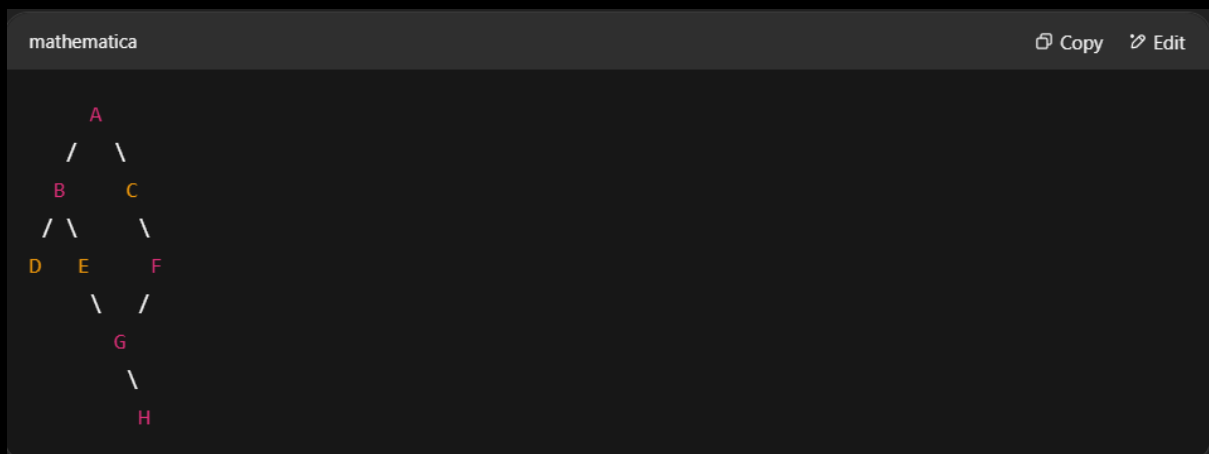
- Two simultaneous BFS-like searches:
 - ♦ One from **start node** → forward
 - ♦ One from **goal node** → backward
- Reduces time complexity from $O(b^d)$ to $O(b^{(d/2)})$
- Requires the graph to be **reversible** (you must be able to search from goal to start)
- Typically uses **two queues and two visited sets**

✅ Bidirectional Search Algorithm Steps:

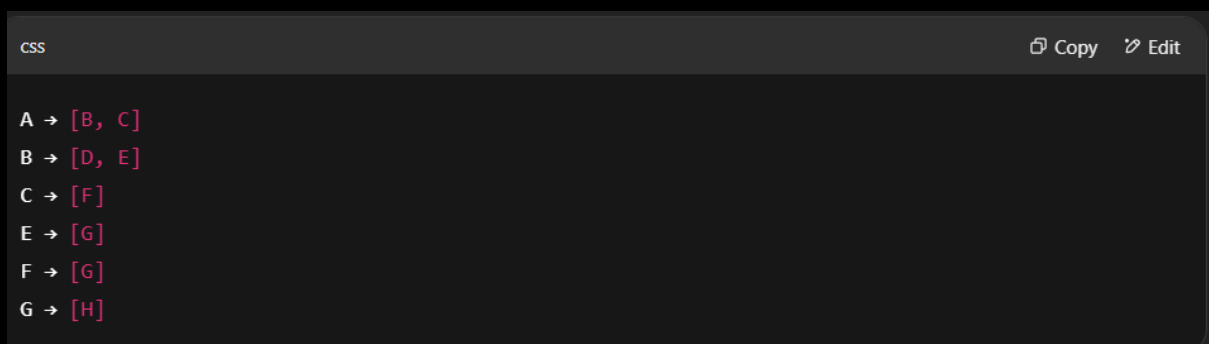
1. Initialize two frontiers:
 - start_queue starting from the start node

- goal_queue starting from the goal node
 - 2. Maintain two visited sets: visited_start, visited_goal
 - 3. Loop:
 - Expand one level from the start_queue
 - Expand one level from the goal_queue
 - If **intersection** of visited sets is found, the path exists
 - 4. Stop when a common node is found (meeting point)
-

Example Graph:



Adjacency List:



Start Node: A

Goal Node: H

Bidirectional Search Execution:

Initial:

- start_queue: [A]
- goal_queue: [H]

Step 1:

- From A $\rightarrow$ add B, C
visited_start: {A, B, C}
- From H $\rightarrow$ no neighbors (reverse search: $H \leftarrow G$)
visited_goal: {H, G}

Step 2:

- From B, C $\rightarrow$ add D, E, F
visited_start: {A, B, C, D, E, F}
- From G (reverse) $\rightarrow$ E, F
visited_goal: {H, G, E, F}

🔗 Now: $\text{visited_start} \cap \text{visited_goal} = \{E, F\} \rightarrow$ ✅ Searches meet!

🔗 Final Path:

One possible path from **A to H** is:

$A \rightarrow B \rightarrow E \rightarrow G \rightarrow H$

or

$A \rightarrow C \rightarrow F \rightarrow G \rightarrow H$ (both valid)

📊 Time and Space Complexity:

Parameter	BFS	Bidirectional Search
Time	$O(b^d)$	$O(b^{d/2})$
Space	$O(b^d)$	$O(b^{d/2})$

b = branching factor, d = depth of solution

🔗 Applications of Bidirectional Search:

- Route finding (e.g., maps, GPS)
- Puzzle solving (e.g., 8-puzzle)
- Social networking (shortest connection path between users)

Let me know if you want a **code implementation** of this in Python or another language!

11. What are Toy Example Problems? Explain any one Toy Problem in detail.

◆ What are Toy Example Problems?

Toy example problems are simple, small-scale problems designed to illustrate or test algorithms, concepts, or techniques in an easy-to-understand way. They are usually:

- **Simplified** versions of more complex real-world problems.
 - Used for **learning**, **demonstration**, and **debugging** algorithms.
 - Helpful for understanding how an algorithm behaves before applying it to larger or more complicated cases.
-

◆ Purpose of Toy Problems:

- Teach algorithm design and problem-solving skills.
 - Provide a clear and focused scenario.
 - Help analyze correctness and efficiency.
 - Test implementation without heavy computational cost.
-

◆ Example Toy Problem: The 8-Puzzle Problem

What is the 8-Puzzle?

The **8-puzzle** is a classic toy problem in artificial intelligence and search algorithms.

- It consists of a **3x3 grid** with **8 numbered tiles** and one empty space.
 - The objective is to move the tiles by sliding them into the empty space to reach a specified goal arrangement.
 - You can only move tiles adjacent to the empty space.
 - It's a good example to demonstrate search techniques like **BFS**, **DFS**, **A***, etc.
-

Problem Setup:

Initial State (start):

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1 2 3

4 _ 6

7 5 8

Goal State:

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1 2 3

4 5 6

7 8 _

Here, _ represents the empty space.

How the problem works:

- At each step, move one tile adjacent to the empty space into that space.
 - The challenge is to find a sequence of moves that transforms the initial state into the goal state.
-

Solving the 8-Puzzle:

- Search algorithms explore the states generated by sliding tiles.
 - Each state represents a node in the search tree.
 - The solution is the path from the initial node to the goal node.
-

Why is the 8-Puzzle a Good Toy Problem?

- Small enough to solve on paper or with simple code.
 - Demonstrates key AI concepts like state-space search, heuristics, and optimization.
 - Has well-defined rules and clear goals.
 - Introduces complexity through the branching factor and number of possible states.
-

If you'd like, I can explain how to solve the 8-puzzle using a specific algorithm like A* or BFS!



12. Vacuum World Problem: Explanation

The **Vacuum World Problem** is a classic **toy problem** in the field of **Artificial Intelligence (AI)**, often used to illustrate basic concepts of agents, perception, action, and goal formulation. It's simple but useful for understanding how an intelligent agent can perceive its environment and make decisions.

Problem Definition:

- The environment consists of **two rooms**: Room A and Room B.
- Each room can be either **clean** or **dirty**.
- A **vacuum agent** is placed in one of the rooms.
- The agent's **goal** is to **clean both rooms**.

State Space:

- Each state consists of:
 - The **location** of the agent: A or B.
 - The **status** of Room A: clean or dirty.
 - The **status** of Room B: clean or dirty.

So the total number of possible states =

$2 \text{ (locations)} \times 2 \text{ (A: clean/dirty)} \times 2 \text{ (B: clean/dirty)} = 8 \text{ states}$

Actions Available to Agent:

The vacuum agent can perform the following actions:

1. **Suck** – Cleans the current room.
2. **Move Left** – Moves to Room A (if in Room B).
3. **Move Right** – Moves to Room B (if in Room A).

Percepts (Input):

The agent can perceive:

- Its **current location** (Room A or B),
 - The **cleanliness status** of the current room.
-

Goal:

- Both rooms must be **clean**.
 - Once both rooms are clean, the task is considered **complete**.
-

Example:

Let's consider an initial state:

- Agent is in **Room A**.
- **Room A** is **dirty**.
- **Room B** is **dirty**.

Sequence of actions:

1. **Suck** (clean Room A)
2. **Move Right**
3. **Suck** (clean Room B)

Now both rooms are clean — **Goal achieved!**

Key AI Concepts Illustrated:

- **State Representation**
 - **Perception-Action Cycle**
 - **Goal-based Agent**
 - **Search and Planning** (in more advanced forms)
-

Uses in AI:

- Used in introductory AI courses to explain:
 - Simple **reflex agents**
 - **Goal-based agents**

- **Search algorithms** for decision-making
-

Let me know if you want this explained with diagrams or if you'd like to see how a reflex agent algorithm works for this problem.

13. Explain any two real world example problems

✓ 1. Traveling Salesperson Problem (TSP)

1. **Problem:**

A salesperson must visit a list of cities, **each exactly once**, and return to the **starting city**.

2. **Goal:**

Find the **shortest possible route** that covers all cities and returns to the origin.

3. **Constraints:**

- Each city visited only once.
- Must return to the starting point.

4. **Applications:**

- Delivery services (e.g., Amazon, courier routes)
- Supply chain logistics
- Printed circuit board (PCB) drilling

5. **Challenges:**

- Number of possible routes increases **exponentially** with the number of cities.
 - Classified as an **NP-Hard** problem in computer science.
-

✓ 2. Shortest Path Problem (e.g., Google Maps)

1. **Problem:**

Find the **minimum-cost path** (shortest distance or time) from a **starting point** to a **destination**.

2. **Goal:**

Reach the destination by traveling the **least possible distance or time**.

3. **Constraints:**

- Must follow available paths/roads.

- Consider road conditions, traffic, or travel cost.

4. Applications:

- GPS navigation (Google Maps, Apple Maps)
- Internet data routing (network packets)
- Robot movement and game AI navigation

5. Algorithms Used:

- **Dijkstra's Algorithm** (for shortest path)
 - **A*** (for optimal and heuristic-based pathfinding)
-

Let me know if you want flowcharts or diagrams for better understanding!

█

14.Explain the various components of a node.

In Artificial Intelligence (AI) search algorithms, a **node** represents a specific **state** in the problem space, along with important metadata used by the search process to reach the goal efficiently.

Various Components of a Node:

1. State

- Represents the current configuration or condition of the problem.
- Example: In a vacuum cleaner problem, a state may be: Vacuum at A, A is clean, B is dirty.

2. Parent Node

- The node from which the current node was generated.
- Helps in tracing the path from the initial state to the current state.

3. Action

- The specific action that was taken to move from the parent node to the current node.
- Example: MoveRight or Suck.

4. Path Cost (g(n))

- The cost of the path from the start node to the current node.
- It accumulates the cost of each action taken.

5. Depth

- Number of steps (or actions) from the start node to the current node.
- Useful for depth-limited search algorithms.

6. Step Cost

- The individual cost of moving from the parent to the current node using the selected action.
- Often used to compute the total path cost.

7. Heuristic Value ($h(n)$) (*only in informed searches like A**)

- Estimated cost from the current node to the goal.
- Used to guide the search towards the goal more efficiently.

8. Evaluation Function ($f(n)$) (*only in informed search*)

- Typically computed as $f(n) = g(n) + h(n)$ in A* search.
- Helps prioritize which node to expand next.

Example (A* Search):

Suppose you're finding a path in a maze:

- **State:** Player at (3,4)
- **Parent:** Node where the player was at (3,3)
- **Action:** Move Right
- **Step Cost:** 1
- **Path Cost $g(n)$:** 5
- **Heuristic $h(n)$:** 3 (estimated steps to goal)
- **Evaluation $f(n)$:** 8

These components are essential for the search algorithm to explore the state space systematically and find an optimal or satisfactory solution.

1

15. Write and explain tree search algorithm.

A **Tree Search Algorithm** is a fundamental algorithm used in Artificial Intelligence (AI) for exploring the possible states in a problem space systematically in the form of a **search tree**. It is a general framework that underlies many specific search strategies like BFS, DFS, A*, etc.

✅ Tree Search Algorithm – Pseudocode

Tree-Search(problem):

Initialize the frontier using the initial state of the problem

Loop do:

 If the frontier is empty, then return failure

 Choose a node from the frontier based on the search strategy

 If the node contains a goal state, then return the solution

 Expand the node, and add the resulting nodes to the frontier

🔍 Explanation of Steps:

1. Initialization

- The algorithm starts with an **initial state** provided by the problem definition.
- This state is inserted into the **frontier** (a list of nodes to be explored).

2. Loop

- The algorithm repeatedly chooses a node from the frontier:
 - In **BFS**, it chooses the shallowest node (FIFO).
 - In **DFS**, it chooses the deepest node (LIFO).
 - In **A***, it chooses the node with the lowest $f(n) = g(n) + h(n)$.

3. Goal Test

- After selecting a node, it checks if it satisfies the **goal test**.
- If yes, it returns the solution (path from start to goal).

4. Expand the Node

- The node is **expanded** by applying all possible actions to generate **successor nodes** (children).
- These children are added to the **frontier**.

5. Termination

- The process repeats until:
 - A goal is found, or
 - The frontier becomes empty (no solution exists).

Key Features:

Feature	Description
---------	-------------

Search Tree	Each node represents a state; edges represent actions
-------------	---

Frontier	Queue/stack/priority queue that manages nodes to explore
----------	--

Expansion	Generation of child nodes from a current node
-----------	---

Goal Test	Determines if a node's state is the goal
-----------	--

Example: Finding a route

Problem: Find a route from city A to city G.

Tree search starts at A and expands possible cities (states) like B, C, D, etc., building a tree of choices until it finds a path to G. The way it explores the tree depends on the specific strategy (BFS, DFS, etc.).

Note:

- Tree search **does not check for repeated states**, so it may **revisit the same state multiple times**, causing inefficiency.
 - In contrast, **graph search** improves this by keeping a list of explored states.
-

Let me know if you want code or an example with BFS/DFS using tree search!

|

16.Explain Breadth First Search. Also discuss its performance.

Refer question 5

|

17. How to Avoid Repeated States & General Graph Search Algorithm

How to Avoid Repeated States

In AI search problems, especially in large or cyclic state spaces (like puzzles or maps), algorithms may visit the **same state multiple times**, leading to:

- **Redundant computation**
- **Infinite loops** (in cyclic graphs)
- **Wasted memory and time**

To avoid this, we use:

Explored Set (Closed List)

- A set or list that stores **already visited states**.
- Before expanding a node, check if its state is in the explored set.
- If yes, **skip it**.
- If not, **add it** to the explored set and expand.

Check before adding to frontier

- Also avoid adding a state to the frontier **if it's already there** with a lower cost.

Example: Repeated States Problem

In the **8-puzzle**, if the blank tile moves left and then right, you return to the same state. Without checking for repetitions, the algorithm would **loop back and forth infinitely**.

General Graph Search Algorithm (with Repeated State Checking)

Graph-Search(problem):

Initialize the frontier using the initial state of the problem

Initialize the explored set to be empty

Loop do:

 If the frontier is empty, return failure

Choose a node from the frontier

If node.state is the goal, return the solution

Add node.state to explored set

For each action in problem.actions(node.state):

 child ← child node from applying action to node.state

 If child.state is not in explored or frontier:

 Add child to the frontier

 Else if child.state is in frontier with higher cost:

 Replace it with the lower-cost child

Explanation of Steps:

1. Frontier:

- Stores the nodes to be explored (queue/stack/priority queue based on the strategy).

2. Explored Set:

- Stores all visited states to prevent revisiting.

3. Node Selection:

- Pick a node from the frontier based on the strategy.

4. Goal Test:

- If the node is the goal, return the path/solution.

5. Expansion:

- Apply all possible actions to generate children.

6. Avoid Duplicates:

- Check if the child's state is already **explored or in frontier**.
- If not, add it to the frontier.

✓ Advantages of Graph Search (Over Tree Search)

Tree Search	Graph Search
May revisit same states	Avoids repeated states
Can fall into infinite loops	Handles cyclic graphs safely
No memory of visited states	Uses an explored set

Let me know if you'd like a real example using BFS or A* that applies graph search!

|

18. Write a note on conformant problem (Sensorless problems)

Conformant Problem (Sensorless Problem)

In Artificial Intelligence, **conformant problems**, also known as **sensorless problems**, are a type of search problem where the agent has **no access to sensory feedback** during its actions. In other words, the agent cannot perceive the current state of the environment while trying to achieve its goal.

🔍 Key Characteristics:

1. **No Observations:** The agent does not receive any information about its current state after taking an action.
 2. **Uncertainty:** The agent must plan while considering all possible states it could be in, due to lack of perception.
 3. **Belief States:** Instead of tracking a single state, the agent tracks a **set of possible states** it might be in, called a **belief state**.
 4. **Goal:** The objective is to **guarantee goal achievement** regardless of the actual initial state or the intermediate unobserved states.
-

🧠 Example:

Imagine a **robot vacuum** in a room where:

- It doesn't have sensors to know its position.
- The room has several obstacles.

- The robot starts in an unknown position.

The robot must plan a sequence of actions (like move left, right, up, down) that will **guarantee** it ends up cleaning the room or reaching a charging station, **no matter where it started** or where it is now.

Conformant Planning Approach:

- The agent represents all possible initial states.
 - After each action, it computes the **resulting belief state**.
 - It continues until all possible resulting states after a sequence of actions satisfy the **goal condition**.
-

Use Cases:

- Robotics in uncertain environments
 - Autonomous systems with limited sensing capabilities
 - Game AI where state information is hidden
-

Conclusion:

Conformant problems are challenging because they require the agent to act **intelligently without sensing**, making planning under **uncertainty** crucial. Solving such problems involves maintaining and updating **belief states** and ensuring that a sequence of actions **leads to success from all possible situations**, even in total darkness.

19.Explain Greedy Best First Search. Consider the map of Romania and the hSLD value table given below. Use greedy best-first search to find a path from Arad to Bucharest.

(THIS ANSWER IS NOT EXACTLY CORRECT BECAUSE IN QUESTION THEY HAVE GIVEN DIAGRAM WITH)

[SO GO FOR QB ANSWER]

Greedy Best-First Search (GBFS) – Explanation with Example

What is Greedy Best-First Search?

Greedy Best-First Search (GBFS) is a search algorithm that expands the node which appears to be **closest to the goal**, using a **heuristic function**.

- It uses:

$$f(n)=h(n)f(n) = h(n)$$

where

- $h(n)$ $h(n)$ = estimated cost from node n to the goal (heuristic)
- GBFS **does not consider** the cost from the start node. It only considers how close it *thinks* it is to the goal.

✚ Romania Map Problem Setup

Let's consider the **Romania map** problem. We want to find the path from **Arad** to **Bucharest** using GBFS.

📄 Heuristic Table (hSLD to Bucharest)

City	hSLD (Straight Line Distance to Bucharest)
Arad	366
Sibiu	253
Fagaras	176
Rimnicu Vilcea	193
Pitesti	100
Bucharest	0
Timisoara	329
Zerind	374
Oradea	380
Lugoj	244
Mehadia	241
Drobeta	242
Craiova	160

City	hSLD (Straight Line Distance to Bucharest)
Giurgiu	77
Urziceni	80
Hirsova	151
Eforie	161
Vaslui	199
Iasi	226
Neamt	234

Steps: Greedy Best-First Search from Arad to Bucharest

Initial State: Start at Arad

- $h(\text{Arad}) = 366$

Step 1: Expand Arad

- Neighbors: **Sibiu (253)**, Timisoara (329), Zerind (374)
- Choose the node with lowest $h \rightarrow$ **Sibiu**

Step 2: Expand Sibiu

- $h(\text{Sibiu}) = 253$
- Neighbors: **Arad**, Fagaras (176), Oradea (380), Rimnicu Vilcea (193)
- Ignore Arad (already visited)
- Lowest h : **Fagaras (176)**

Step 3: Expand Fagaras

- $h(\text{Fagaras}) = 176$
- Neighbors: **Sibiu**, **Bucharest (0)**
- Choose **Bucharest**

 Goal reached.

 **Final Path Found by GBFS:**

Arad $\rightarrow$ Sibiu $\rightarrow$ Fagaras $\rightarrow$ Bucharest

⚠ Note:

- GBFS is **fast and memory efficient**, but **not optimal** or complete.
- It can get **trapped in local minima** because it only follows the heuristic and ignores actual cost.

Let me know if you'd like a visual of the path or a comparison with A* Search!

20.Explain A* algorithm. Consider the map of Romania and the hSLD value table given. Use A* algorithm to find a path from Arad to Bucharest.

✅ A Search Algorithm – Explanation with Example*

🔍 What is A Algorithm?*

A* is an informed search algorithm that combines the advantages of **Uniform Cost Search** and **Greedy Best-First Search**.

It uses the evaluation function:

$$f(n)=g(n)+h(n)f(n) = g(n) + h(n)$$

Where:

- $g(n)$: cost from the start node to the current node
- $h(n)$: estimated cost from the current node to the goal (heuristic)
- $f(n)$: estimated total cost of the path through the current node to the goal

🎯 Goal: Find the optimal path from Arad to Bucharest

📖 Map: Romania

📄 Heuristic Table (Straight Line Distance to Bucharest)

City	hSLD to Bucharest
Arad	366
Sibiu	253

City	hSLD to Bucharest
Fagaras	176
Rimnicu Vilcea	193
Pitesti	100
Bucharest	0
Timisoara	329
Zerind	374
Oradea	380
Lugoj	244
Mehadia	241
Drobeta	242
Craiova	160
Giurgiu	77
Urziceni	80

Step-by-Step Execution

Start at Arad

- $g(\text{Arad}) = 0$
- $h(\text{Arad}) = 366$
- $f(\text{Arad}) = 0 + 366 = \mathbf{366}$

Expand Arad → Neighbors:

- Zerind: $g = 75$, $h = 374$, $f = \mathbf{449}$
- Sibiu: $g = 140$, $h = 253$, $f = \mathbf{393}$
- Timisoara: $g = 118$, $h = 329$, $f = \mathbf{447}$

 Pick **Sibiu** (lowest f)

Expand Sibiu → Neighbors:

- Arad (already visited)
- Fagaras: $g = 140 + 99 = 239$, $h = 176$, $f = 415$
- Rimnicu Vilcea: $g = 140 + 80 = 220$, $h = 193$, $f = 413$
- Oradea: $g = 140 + 151 = 291$, $h = 380$, $f = 671$

✓ Pick **Rimnicu Vilcea**

🔄 **Expand Rimnicu Vilcea → Neighbors:**

- Sibiu (already visited)
- Pitesti: $g = 220 + 97 = 317$, $h = 100$, $f = 417$
- Craiova: $g = 220 + 146 = 366$, $h = 160$, $f = 526$

✓ Pick **Pitesti**

🔄 **Expand Pitesti → Neighbors:**

- Rimnicu Vilcea (visited)
- Craiova
- Bucharest: $g = 317 + 101 = 418$, $h = 0$, $f = 418$

✓ Pick **Bucharest** — GOAL REACHED 🎯

✓ ***Final Path (A)**:**

Arad → Sibiu → Rimnicu Vilcea → Pitesti → Bucharest

📊 **Total Cost:**

$140(\text{Arad} \rightarrow \text{Sibiu}) + 80(\text{Sibiu} \rightarrow \text{Rimnicu Vilcea}) + 97(\text{RV} \rightarrow \text{Pitesti}) + 101(\text{Pitesti} \rightarrow \text{Bucharest}) = **418**$
 $140(\text{Arad} \rightarrow \text{Sibiu}) + 80(\text{Sibiu} \rightarrow \text{Rimnicu Vilcea}) + 97(\text{RV} \rightarrow \text{Pitesti}) + 101(\text{Pitesti} \rightarrow \text{Bucharest}) = **418**$

⚖️ **Why A* is Better?**

- **Optimal:** Always finds the shortest path (if $h(n)$ is admissible)
- **Complete:** Finds a solution if one exists
- **Efficient:** Uses both actual and estimated costs

Let me know if you'd like a visualization of the Romania map with this path!

21. Explain Recursive Best First Search algorithm. Consider the map of Romania and the hSLD value table given. Use RBFS* algorithm to find a path from Arad to Bucharest.

21. Recursive Best-First Search (RBFS) Algorithm

What is Recursive Best-First Search (RBFS)?

RBFS (Recursive Best-First Search) is an informed search algorithm similar to A*, but uses **less memory**.

It combines the **heuristic guidance** of A* with the **space-efficiency** of depth-first search.

Key Features of RBFS:

1. Uses **heuristics** to guide the search.
 2. Maintains only a **limited amount of memory** (like DFS).
 3. Backtracks when the current path exceeds the cost limit.
 4. Does **not remember all generated nodes**, so it's **memory efficient**.
-

Algorithm Steps (in pointwise):

1. Start at the initial node (e.g., **Arad**).
 2. Use the heuristic function $f(n) = g(n) + h(n)$:
 - $g(n)$ = cost from start to current node
 - $h(n)$ = estimated cost from node to goal (from hSLD table)
 3. Explore the **best (lowest f-value)** child node.
 4. If the best node's f-value **exceeds the alternative**, backtrack and explore the alternative.
 5. Continue recursively until the **goal (Bucharest)** is found or no paths remain.
-

Apply RBFS on Romania Map (Arad → Bucharest)

Let's assume the following **straight-line distances (hSLD values)** to Bucharest:

City	hSLD (heuristic)
Arad	366
Sibiu	253
Fagaras	176
Rimnicu Vilcea	193
Pitesti	100
Bucharest	0

Step-by-Step Path Using RBFS (Arad → Bucharest):

1. **Start at Arad (h=366)**
 - Possible moves: Sibiu, Timisoara, Zerind
 - Best: **Sibiu (g=140, h=253 → f=393)**
2. **Move to Sibiu (f=393)**
 - Possible moves: Arad (visited), Fagaras, Rimnicu Vilcea, Oradea
 - Best: **Fagaras (g=239, h=176 → f=415)**
3. **Move to Fagaras (f=415)**
 - Possible move: Bucharest (g=418, h=0 → f=418)
4. **Move to Bucharest – Goal Reached**

Final Path Found Using RBFS:

```
nginx
```

```
Copy Edit
```

```
Arad → Sibiu → Fagaras → Bucharest
```

Advantages of RBFS:

- Uses **less memory** than A* (good for large search trees).
- Can find **optimal paths** if heuristic is admissible.

⚠ Disadvantages:

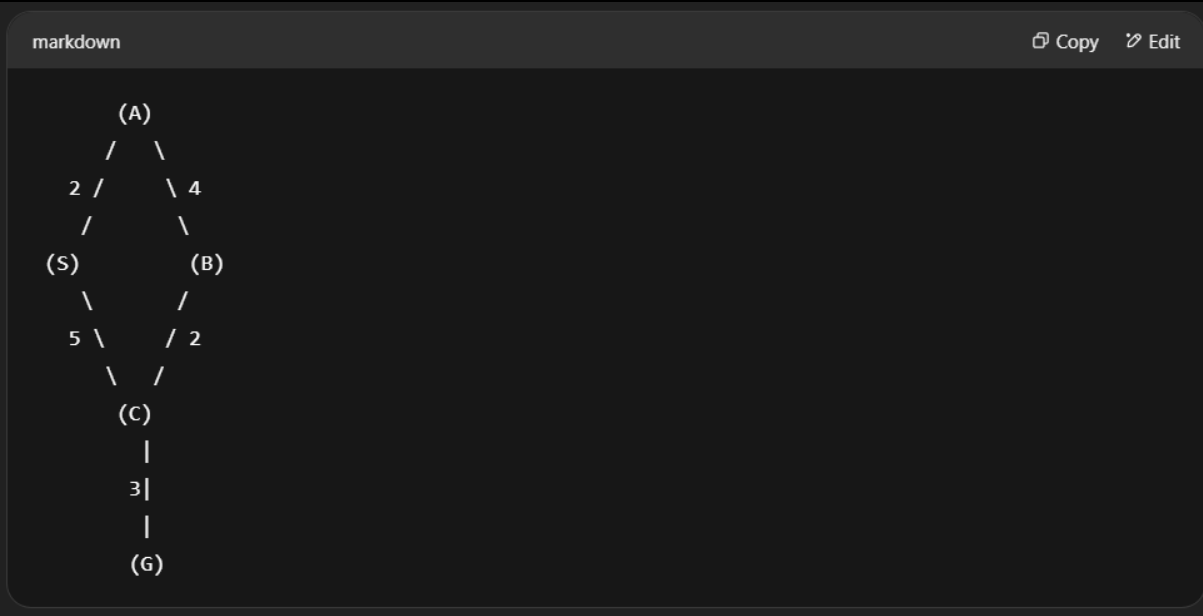
- May do **more work than A*** due to repeated re-expansion.
- Slower than A* in practice, but **better in memory-constrained situations**.

Let me know if you'd like a tree diagram or code implementation of RBFS!

22. Write a note on SMA* algorithm.

✅ SMA* Example: Pathfinding on a Simple Graph

Let's say we want to find the shortest path from Start (S) to Goal (G) in the following graph:



📌 Heuristic (h-values):

Node	$h(n)$
S	6
A	4
B	3
C	2
G	0

 Step-by-Step SMA* (with memory limit of only 4 nodes):

► Start at S:

- $f(S) = g + h = 0 + 6 = 6$
- Expand children: A ($f=6$), B ($f=7$), C ($f=7$)

► Memory: [S, A, B, C] – memory full!

Now expand A (lowest f):

- $A \rightarrow G$ through an unlisted path (not shown), so skip.
- No further children for A. Remove A from memory.

► Now memory has: [S, B, C]

Add next best node – say C has a child G with $g = 5 + 3 = 8$ and $h = 0 \Rightarrow f(G) = 8$

Now memory: [S, B, C, G] – full again.

► Node with worst f is G (8) $\rightarrow$ if needed, it may be removed and re-generated.

But G is the goal $\rightarrow$ so SMA* returns:

$S \rightarrow C \rightarrow G$

with total cost = 5 (S to C) + 3 (C to G) = 8

 What Happens if Memory is Smaller?

If memory was only 3 nodes, SMA* might:

- Remove C or B earlier,
 - Save their f -values for backtracking,
 - Re-expand them later if a better path isn't found.
-

 Conclusion:

This shows how SMA*:

- Expands like A*,
- Keeps memory bounded,
- Removes worst nodes when full,
- Can still find the optimal or near-optimal path depending on memory available.

Let me know if you want this in diagram form too!

23. Write a note on IDA* algorithm.

✓ What is IDA*?

IDA* is a search algorithm that combines the advantages of:

- Iterative Deepening Depth-First Search (IDDFS)
- and A* search

It is used to find the **optimal path** in a search space with **limited memory**, like A*, but more memory-efficient.

🔑 Key Characteristics:

1. Uses $f(n) = g(n) + h(n)$ like A* where:
 - $g(n)$ = cost from start to current node
 - $h(n)$ = heuristic estimate to goal
 2. Instead of storing all nodes in memory, it performs **depth-first searches** with increasing cost limits.
 3. Each iteration sets a **cost threshold** equal to the smallest cost that exceeded the previous threshold.
 4. It **prunes paths** that exceed the current cost limit.
-

📝 Algorithm Steps (Pointwise):

1. Initialize the **threshold** with $f(\text{start})$.
 2. Perform a **depth-limited DFS** where nodes with $f(n) > \text{threshold}$ are pruned.
 3. If goal found within threshold, return path.
 4. Otherwise, update threshold to the **minimum $f(n)$ that exceeded the threshold** in the previous iteration.
 5. Repeat steps 2–4 until goal is found or no more nodes to explore.
-

🔍 Example: Pathfinding from A to G

Consider the graph:



- Costs: $A \rightarrow B=1$, $A \rightarrow C=2$, $B \rightarrow D=4$, $C \rightarrow G=5$
- Heuristics (h): $B=5$, $C=3$, $D=2$, $G=0$

Stepwise IDA* Search:

- Start at A, threshold = $f(A) = g(A) + h(A) = 0 + h(A)$. Assume $h(A) = 6$.
- Iteration 1 (threshold=6):**
Explore paths with $f \leq 6$:
 - $A \rightarrow B$: $g=1$, $h=5 \rightarrow f=6$ (allowed)
 - $A \rightarrow C$: $g=2$, $h=3 \rightarrow f=5$ (allowed)
 - $A \rightarrow B \rightarrow D$: $g=5$, $h=2 \rightarrow f=7$ (exceeds threshold, prune)
 - $A \rightarrow C \rightarrow G$: $g=7$, $h=0 \rightarrow f=7$ (exceeds threshold, prune)
- No goal found, update threshold to min exceeded $f = 7$.
- Iteration 2 (threshold=7):**
Explore paths with $f \leq 7$:
 - $A \rightarrow B \rightarrow D$: $f=7$ (allowed)
 - $A \rightarrow C \rightarrow G$: $f=7$ (allowed)

Goal G found via path: $A \rightarrow C \rightarrow G$ with cost 7.

Advantages of IDA*:

- Uses **linear memory** like DFS.
- Finds **optimal path** if heuristic is admissible.
- Combines **best-first** search efficiency with **memory efficiency**.

Disadvantages:

- Can revisit nodes multiple times, leading to more computation.
 - More time-consuming than A* in some cases.
-

If you'd like, I can provide code or more detailed examples for IDA*!

24. What are heuristic functions? Explain the two heuristics used in 8 puzzle. Also compute the same for the 8 puzzle instance given below :

[Refer QB answer]

25. Write a note on inventing heuristic functions.

 **25. Note on Inventing Heuristic Functions**

What is a Heuristic Function?

A **heuristic function ($h(n)$)** is a function used in search algorithms (like A*, Greedy Best-First Search) that estimates the **cost or distance from a node n to the goal**.

- It helps the algorithm decide which path to explore first.
 - A **good heuristic** can greatly reduce search time.
-

Key Properties of Heuristic Functions:

1. **Admissible:**
The heuristic **never overestimates** the true cost to reach the goal.
 $h(n) \leq \text{actual cost from } n \text{ to goal}$
 2. **Consistent (Monotonic):**
For every node n and successor n' ,
 $h(n) \leq \text{cost}(n, n') + h(n')$
 3. **Efficient to Compute:**
The heuristic should be fast to calculate.
-

How to Invent Heuristic Functions?

1. **Relax the Problem:**
Simplify the problem by removing some constraints to get an easier version.

- The solution cost to this easier problem can be used as a heuristic.
 - Example: In pathfinding, ignoring obstacles to calculate straight-line distance.
 - 2. **Use Domain Knowledge:**
Use knowledge about the problem to create an estimate.
 - Example: Number of misplaced tiles in the 8-puzzle.
 - 3. **Use Precomputed Data:**
Store known distances or costs between nodes to estimate.
 - 4. **Combine Heuristics:**
Take the maximum or weighted sum of multiple heuristics to improve accuracy while maintaining admissibility.
-

Example: Inventing a Heuristic for 8-Puzzle

- **Problem:** Move tiles to reach the goal configuration.
 - **Heuristic Idea:**
 - **Number of misplaced tiles:** Count how many tiles are not in their goal position.
 - This never overestimates moves needed (admissible).
 - Simple and fast to compute.
 - **Better heuristic:**
 - **Manhattan distance:** Sum of distances of tiles from their goal positions measured by moves (up/down/left/right).
 - Provides a more accurate estimate.
-

Summary:

- Inventing heuristics involves **simplifying** the problem or **using problem knowledge** to estimate cost.
 - Good heuristics improve search efficiency by **guiding the algorithm effectively** without losing optimality.
-

Want me to help you invent heuristics for a specific problem?

1

26. Explain Hill Climbing Search.

26. Hill Climbing Search

What is Hill Climbing Search?

Hill Climbing is a **local search algorithm** that continuously moves towards a better solution by choosing the best neighboring state at each step.

- It tries to **maximize (or minimize)** an objective function (called evaluation or heuristic function).
 - It's called "hill climbing" because it imagines the solution space as a landscape of hills and valleys, and it keeps moving uphill towards the peak (best solution).
-

Key Characteristics:

1. Starts with an **initial state**.
 2. Examines the **neighbors** of the current state.
 3. Moves to the **neighbor with the best improvement**.
 4. Stops when no neighbor improves the current state (local maximum reached).
 5. It is **greedy** and only considers immediate neighbors.
-

Limitations:

- Can get stuck in **local maxima** (not the best global solution).
 - Can get stuck on **plateaus** (flat areas with no better neighbor).
 - Can get stuck in **ridges** (narrow peaks difficult to climb).
-

Example: Finding Maximum of a Function

Suppose you want to find the maximum of this function (discrete values):

x	f(x)
0	1
1	3
2	7
3	8
4	6
5	5

Hill Climbing Steps:

- Start at $x = 1$ ($f=3$).
- Check neighbors:
 - $x=0 \rightarrow f=1$ (worse)
 - $x=2 \rightarrow f=7$ (better)
- Move to $x=2$.
- Check neighbors:
 - $x=1 \rightarrow f=3$ (worse)
 - $x=3 \rightarrow f=8$ (better)
- Move to $x=3$.
- Check neighbors:
 - $x=2 \rightarrow f=7$ (worse)
 - $x=4 \rightarrow f=6$ (worse)
- No better neighbor, stop here.

Result:

Hill Climbing finds $x=3$ with value **8** as a local maximum.

Summary:

- Hill Climbing is a simple **greedy local search** algorithm.

- Useful for problems where you can move stepwise to better neighbors.
 - Risk of **getting stuck** in local maxima or plateaus.
-

Want me to explain variants like **Stochastic Hill Climbing** or **Simulated Annealing**?

27. Explain the issues due to which hill climbing gets stuck.

27. Issues Due to Which Hill Climbing Gets Stuck

Hill Climbing is a simple search algorithm, but it has several common problems that can cause it to **fail to find the global optimum**:

1. Local Maxima (or Local Optima)

- **What:** The algorithm reaches a state better than all its neighbors but not the best overall (global maximum).
- **Why it's a problem:** It stops, thinking it found the best solution, but there is a better one elsewhere.

Example:

Imagine a mountain range with many peaks. The algorithm reaches a smaller peak (local maximum) and stops, missing the tallest peak (global maximum).

x	f(x)
0	1
1	3
2	5
3	4
4	6
5	2

If started at $x=0$, it may climb to $x=2$ (local max = 5) and stop, missing the global max at $x=4$ (value 6).

3. Plateau

- **What:** A flat area in the search space where neighboring states have the same value.

- **Why it's a problem:** The algorithm has no direction to move since no neighbor is better.

Example:

x	f(x)
0	5
1	5
2	5
3	6

At $x=1$ or $x=2$, the algorithm is stuck on a plateau with no better neighbor, even though $x=3$ has a better value.

4. Ridges

- **What:** Narrow, curved paths leading to better solutions, but the immediate neighbors may appear worse.
- **Why it's a problem:** The algorithm may fail to move along the ridge because it only considers immediate neighbors and can't "jump" to better states.

Example:

A steep, narrow ridge on a mountain that requires moving sideways first (which looks worse) before climbing up.

5. Step Size Problem

- **What:** If the search moves in fixed-sized steps, it might miss the optimum because the step size is too big or too small.

Summary of Issues

Summary of Issues

Issue	Description	Example
Local Maxima	Stuck at a peak that is not the highest	Mountain range with smaller peak
Plateau	Stuck on flat area with equal neighbors	Flat region with no better neighbors
Ridge	Cannot follow narrow path to peak	Narrow, curved mountain ridge
Step Size	Step too large or small to find optimum	Missed peak due to step jumps

28. Briefly Explain Simulated Annealing algorithm.

28. Simulated Annealing Algorithm

What is Simulated Annealing?

Simulated Annealing (SA) is a **probabilistic optimization algorithm** inspired by the **annealing process in metallurgy**, where materials are heated and then slowly cooled to reach a low-energy (optimal) state.

- SA tries to **avoid getting stuck in local optima** by occasionally accepting worse solutions.
- It uses a **temperature parameter (T)** that gradually decreases during the process.

How Simulated Annealing Works (Stepwise):

1. **Start** with an initial solution and an initial high temperature T .
2. **Repeat** until stopping condition:
 - Generate a **neighbor solution**.
 - Calculate the **change in cost** $\Delta E = \text{cost}(\text{new}) - \text{cost}(\text{current})$.
 - If $\Delta E < 0$ (better solution), **accept** the new solution.
 - Else, accept the new solution with probability:

$$P = e^{-\Delta E/T}$$

- **Reduce the temperature T** gradually (cooling schedule).
3. When temperature is low enough, **stop** and return the current solution.

Example: Finding Minimum of $f(x) = x^2$

- Start at $x = 10$ (cost = 100).
- Initial temperature $T = 100$.
- Generate neighbor, e.g., $x = 9$ (cost = 81), better $\rightarrow$ accept.
- Generate neighbor, e.g., $x = 12$ (cost = 144), worse $\rightarrow$ accept with probability $e^{-(144-81)/T}$.
- Gradually decrease T .
- Over iterations, moves concentrate around $x=0$ (minimum).

Advantages:

- Can **escape local minima** by accepting worse solutions.
- Simple to implement.
- Works well for large, complex problems.

Disadvantages:

- Requires careful **cooling schedule** design.
- May take longer to converge.

29. Compare various uninformed searching technique.

Comparison of Various Uninformed Search Techniques

Uninformed (or blind) search algorithms explore the search space without any domain-specific knowledge (no heuristics). Here's a comparison of common uninformed search methods:

Search Algorithm	Completeness	Time Complexity	Space Complexity	Optimality	Description / Notes	📄
Breadth-First Search (BFS)	Yes (if branching factor is finite)	$O(b^d)$ where b = branching factor, d = depth of shallowest goal	$O(b^d)$	Yes (if all step costs are equal)	Explores nodes level by level; uses a queue; guaranteed shortest path in uniform cost	
Depth-First Search (DFS)	No (may get stuck in infinite depth)	$O(b^m)$ where m = max depth of search tree	$O(bm)$	No	Explores as deep as possible before backtracking; uses a stack; low memory usage	
Depth-Limited Search (DLS)	No (may miss goal beyond limit)	$O(b^l)$ where l = depth limit	$O(bl)$	No	DFS with a depth cutoff; prevents infinite search	
Iterative Deepening DFS (IDDFS)	Yes	$O(b^d)$	$O(bd)$	Yes (if step costs equal)	Repeatedly applies DLS with increasing depth; combines DFS space efficiency with BFS completeness	
Uniform Cost Search (UCS)	Yes	$O(b^{\lceil C^*/\epsilon \rceil})$ where C^* = cost of optimal solution, ϵ = min step cost	$O(b^{\lceil C^*/\epsilon \rceil})$	Yes	Expands least cost node first; optimal for varying step costs; uses priority queue	

Summary Points:

- **BFS** is complete and optimal (for equal step costs) but uses lots of memory.
- **DFS** is memory efficient but not guaranteed to find a solution or optimal solution.
- **DLS** avoids infinite loops but can miss solutions deeper than the limit.
- **IDDFS** combines advantages of BFS and DFS, complete and optimal, with less memory than BFS.
- **UCS** is optimal for variable costs but can be slower than BFS and uses significant memory.

30.Explain genetic algorithm with example.

[Better you read qb answer. Cz its algorithm]

🧠 Genetic Algorithm (GA)

✅ What is a Genetic Algorithm?

Genetic Algorithm is a **search and optimization technique** inspired by the process of **natural selection and genetics** in biology.

- It is used to find approximate solutions to optimization and search problems.
- It works with a **population** of candidate solutions called **chromosomes**.
- Solutions evolve over generations using operators like **selection, crossover, and mutation**.

Basic Concepts:

- **Chromosome:** Representation of a candidate solution (often a string of bits or numbers).
 - **Population:** Set of all candidate solutions at a given iteration.
 - **Fitness Function:** Measures how good each solution is.
 - **Selection:** Choosing the best chromosomes to reproduce.
 - **Crossover (Recombination):** Combining parts of two chromosomes to create offspring.
 - **Mutation:** Randomly changing some parts of chromosomes to maintain diversity.
-

Genetic Algorithm Steps:

1. **Initialize** a random population of chromosomes.
 2. **Evaluate** fitness of each chromosome.
 3. **Select** pairs of chromosomes for reproduction based on fitness.
 4. **Crossover** pairs to produce new offspring.
 5. **Mutate** offspring randomly.
 6. **Replace** old population with new offspring.
 7. Repeat steps 2-6 until stopping criteria (e.g., number of generations or satisfactory fitness) are met.
-

Example: Solving Maximize $f(x) = x^2$ for x in $[0, 31]$

- Represent x as 5-bit binary string (chromosome), e.g., 10101 = 21.
 - Fitness = x^2 .
 - Initial population: Random 5-bit strings.
 - Select chromosomes with higher fitness.
 - Perform crossover and mutation.
 - Over generations, population evolves towards binary strings representing numbers close to 31 (max fitness $31^2 = 961$).
-

Advantages:

- Good for large, complex, or poorly understood search spaces.
 - Can find near-optimal solutions.
 - Does not require gradient or differentiability.
-

⚠ Disadvantages:

- Can be slow.
- Requires careful tuning of parameters (mutation rate, population size).
- May converge prematurely to suboptimal solutions.

|

31. Explain local beam search with example

🧠 Local Beam Search

✅ What is Local Beam Search?

Local Beam Search is a **search algorithm** that keeps track of **k states (called beams)** instead of just one, and explores neighbors of all these states at each step.

- Unlike Hill Climbing which moves from a single state to the best neighbor, Local Beam Search keeps multiple candidates to avoid getting stuck.
 - It selects the best **k successors** from all neighbors of the current beams for the next iteration.
-

🔑 How Local Beam Search Works:

1. Start with **k randomly generated states** (the beam).
 2. Generate all successors of these k states.
 3. From all these successors, select the **k best** states based on the evaluation function.
 4. Repeat steps 2 and 3 until a goal state is found or no improvement occurs.
-

🔍 Example:

Suppose we want to find the maximum value of a function.

- **k = 2** (keep track of 2 states)

- Initial states: $x = 1$ ($f=3$), $x = 4$ ($f=5$)
- Successors:
 - For $x=1$: neighbors $x=0$ ($f=1$), $x=2$ ($f=7$)
 - For $x=4$: neighbors $x=3$ ($f=6$), $x=5$ ($f=5$)
- Combine neighbors: $\{0(1), 2(7), 3(6), 5(5)\}$
- Select top 2: $x=2$ ($f=7$), $x=3$ ($f=6$)
- Next iteration, generate successors of $x=2$ and $x=3$, continue similarly.

✓ Advantages:

- Maintains diversity by keeping multiple states.
- Less likely to get stuck in local maxima than Hill Climbing.
- Faster convergence than random restart.

⚠ Disadvantages:

- If all beams converge to the same local maximum, it behaves like Hill Climbing.
- Requires more memory to keep track of k states.

```

function SIMULATED-ANNEALING(problem, schedule) returns a solution state
  inputs: problem, a problem
           schedule, a mapping from time to "temperature"
  local variables: current, a node
                    next, a node
                    T, a "temperature" controlling the probability of downward steps

  current  $\leftarrow$  MAKE-NODE(INITIAL-STATE[problem])
  for  $t \leftarrow 1$  to  $\infty$  do
    T  $\leftarrow$  schedule[t]
    if T = 0 then return current
    next  $\leftarrow$  a randomly selected successor of current
     $\Delta E \leftarrow$  VALUE[next] - VALUE[current]
    if  $\Delta E > 0$  then current  $\leftarrow$  next
    else current  $\leftarrow$  next only with probability  $e^{\Delta E/T}$ 

```

[ALGORITHM IMAGE FROM QB]